

## RESEARCH/WORK EXPERIENCE

### UNIVERSITY OF CAMBRIDGE

#### Research Collaborator - Machine Learning

October 2026 - Present

Remote

Cambridge, United Kingdom

- Conducting research with Prof. Mateja Jamnik and Xiangjian Jiang on robust and scalable prototype-based local feature selection for tabular data, extending the ProtoGate framework.

### MILA - QUEBEC AI INSTITUTE

#### Research Collaborator - Machine Learning

September 2026 - Present

Montreal, Canada

- Conducting an ongoing research project with Prof. Guy Wolf and Adrien Aumon on unsupervised forests for tabular compression.

### INRIA, SODA TEAM

#### Machine Learning Research Assistant

September 2025 - Present

(full-time through January 2026, part-time/remote thereafter)

Palaiseau, France

- **First author "Disentangling Scale, Diversity, and Source in Pan-Cancer Survival Prediction Using TCGA and SEER", in preparation for submission to GigaScience, with Gaël Varoquaux and Judith Abécassis (INRIA SODA)**
- Quantifying the efficacy of pan-cancer vs. single-cancer training by harmonizing the TCGA (11k patients) and SEER (7M patients) cohorts, creating one of the largest unified clinical datasets for survival prediction
- Utilized advanced tabular representation learning (SurvivalBoost, TARTE, sentence embeddings) to evaluate model transferability across 31 shared clinical features
- Developed a robust data pipeline to resolve cross-cohort discrepancies, including sentinel normalization, TNM derivation, and patient-level deduplication

### BAR-ILAN UNIVERSITY

#### Machine Learning Research Assistant

June - September 2026

Ramat Gan, Israel

- Conducted research with Prof. Ofir Lindenbaum on semi-supervised learning and foundation-model adaptation for tabular data under limited-label regimes.
- Explored and developed methods for extracting useful signal from unlabeled tabular data, including graph-based propagation, iterative pseudo-labelling, representation learning, and geometry-aware regularisation.
- Investigated semi-supervised adaptation strategies for tabular foundation models such as TabPFN and TabICL, incorporating self-training, Laplacian objectives, unlabeled-sample attention, and embedding-similarity constraints.  
<https://github.com/dansuissa/tabular-foundation-ssl>

### EPFL (École Polytechnique Fédérale de Lausanne)

#### Machine Learning Research Assistant

June 2024 - September 2025

(full-time through September 2024, part-time/remote thereafter)

Lausanne, Switzerland

- Worked under the supervision of Professor Pascal Fua, Romuald Houdré and Nicolas Villa in the Laboratory of Quantum and Nano-Optics (LQNO)
- Pioneered the application of time-series and tabular ML classifiers (XGBoost, MultiRocket, HIVE-COTE 2.0) for antimicrobial susceptibility testing, representing the first-ever use of these architectures with photonic crystal optical transmission data
- Engineered a custom pipeline to translate raw optical signals into high-accuracy biological sensing predictions : [https://github.com/dansuissa/EPFL\\_AMR](https://github.com/dansuissa/EPFL_AMR)

### INSERM

#### Machine Learning Research Assistant (Bachelor Thesis)

January - March 2025

Paris, France

- Worked under the supervision of Professor Guillaume Fillion, Ariel Lindner and Amir Pandi
- Extended a codon language model, CodonTransformer ([nature.com/articles/](https://nature.com/articles/)), to optimize regulatory DNA sequences using novel tokenization and embedding techniques, enhancing biological sequence encoding :  
<https://github.com/dansuissa/CodonTransformerV2>

### ÉCOLE POLYTECHNIQUE

#### Undergraduate Machine Learning Research Assistant

September - December 2024

Palaiseau, France

- In collaboration with Pr. Davide Buscaldi, evaluated LLM unlearning methods on the SemEval-2025 dataset to reduce memorization of sensitive data beyond output-filtering baselines

EDUCATION

MCGILL UNIVERSITY  
MEng Artificial Intelligence

January 2026 - May 2027  
Montreal, Canada

ECOLE POLYTECHNIQUE  
BSc Mathematics and Computer Science

August 2022 - September 2025  
Palaiseau, France

STANFORD UNIVERSITY  
International Honors Program

June - August 2023  
Stanford CA, USA

PROJECTS

- **Research Project - McGill University (8-month, part-time), supervised by Prof. Di Wu:** Benchmarking the trustworthiness of small (<10B-parameter) large language models across diverse hallucination scenarios, characterizing reliability and failure modes, and investigating training-free mitigation strategies for safer deployment.
- **Theory of Machine Learning Project:** Combined proof-based machine learning theory with empirical analysis to derive and validate scaling laws for overparameterized linear regression and two-layer neural networks, with a focus on SGD dynamics, generalization, and NTK, mean-field, and random-features limits

SKILLS

Python (TensorFlow, scikit-learn, skrub, PyTorch, numpy, pandas), R, C/C++, Git, Linux, Coq

LANGUAGES

- French (Native)
- English (Fluent)
- Spanish (Intermediate)
- Hebrew (Intermediate)

VOLUNTEERING & EXTRACURRICULARS

Scout movement group leader · Maths Olympiads tutor · IANTD Rescue Diver · MMA assistant coach